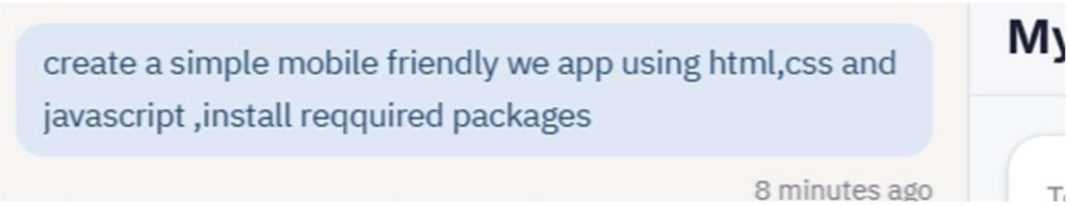
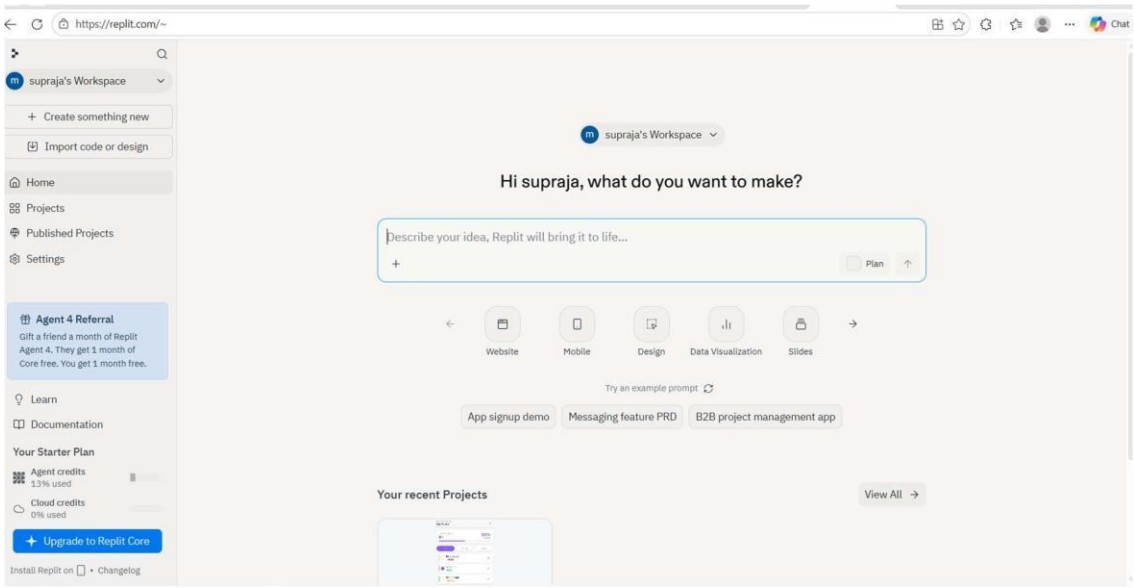
AI & AC
ASSIGNMENT-24.1

HT NO:2303A51644
BATCH:28

Task-1:Environment Setup Prompt:



Setup:



Explanation:

1. **A free account was created in Replit to use it as a cloud-based IDE.**
2. **A new project (Repl) was created by selecting the HTML, CSS, and JavaScript template.**
3. **Different templates like React Native, Flutter, and HTML/CSS/JS were explored for mobile app development.**

4. **The Replit interface, including editor, console, and file structure, was understood.**
5. **The environment was prepared for building and testing mobile applications.**

Task-2: Basic Mobile App

Prompt:

disply app title "Hello App" input filed to enter users name , Button to submit ,show greeting message "Hello,[name]" dynamically and keep it simple and responsive for mobile

Code:

lib > api-client-react > package.json

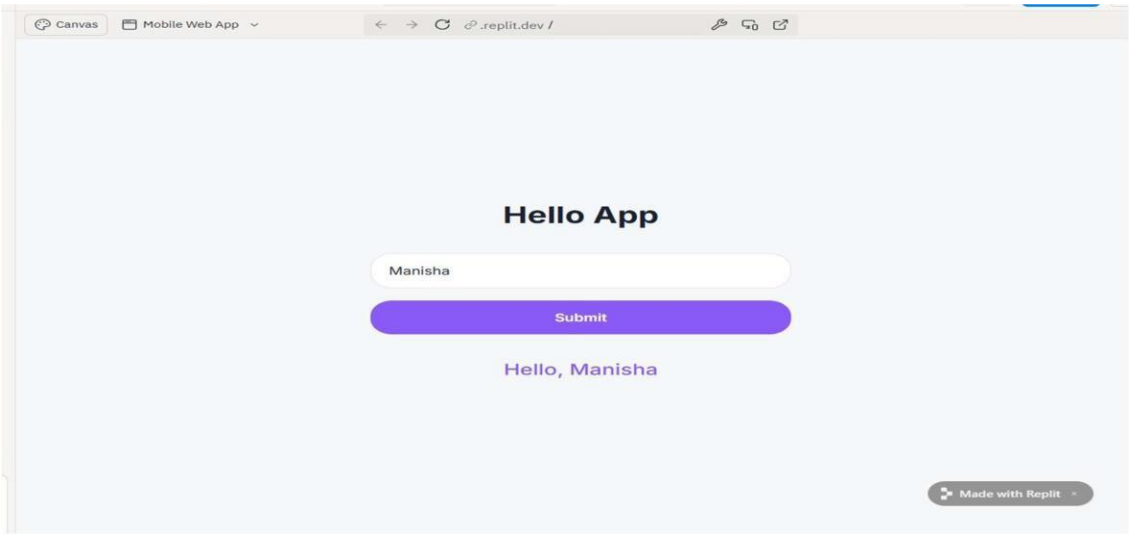
```
{
  "name": "@workspace/api-client-react",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "exports": {
    ".": "./src/index.ts"
  },
  "dependencies": {
    "@tanstack/react-query": "catalog:"
  },
  "peerDependencies": {
    "react": ">=18"
  }
}
```

lib > api-spec > openapi.yaml

```yaml
openapi: 3.1.0
info:
  # Do not change the title, if the title changes, the import paths will be broken
  title: Api
  version: 0.1.0
  description: API specification
servers:
  - url: /api
    description: Base API path
tags:
  - name: health
    description: Health operations
paths:
  /healthz:
    get:
      operationId: healthCheck
      tags: [health]
      summary: Health check
      description: Returns server health status
      responses:
        "200":
          description: Healthy
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/HealthStatus"
components:
  schemas:
    HealthStatus:
      type: object
      properties:
```

```
import { defineConfig, InputTransformerFn } from "orval";
import path from "path";

const root = path.resolve(__dirname, "..", "..");
const apiClientReactSrc = path.resolve(root, "lib", "api-client-react", "src");
const apiZodSrc = path.resolve(root, "lib", "api-zod", "src");

// Our exports make assumptions about the title of the API being "Api" (i.e. generated
const titleTransformer: InputTransformerFn = (config) => {
  config.info ??= {};
  config.info.title = "Api";

  return config;
};

export default defineConfig({
  "api-client-react": {
    input: {
      target: "./openapi.yaml",
      override: {
        transformer: titleTransformer,
      },
    },
    output: {
      workspace: apiClientReactSrc,
      target: "generated",
      client: "react-query",
      mode: "split",
      baseUrl: "/api",
      clean: true,
```

Output:



1. **Explanation:A simple mobile-friendly application called “Hello App” was created using HTML, CSS, and JavaScript in Replit.**
2. **The app includes an input field where the user can enter their name.**
3. **A button is provided to submit the input entered by the user.**
4. **When the button is clicked, a greeting message like “Hello, [name]” is displayed dynamically on the screen.**
5. **AI assistance in Replit was used to generate the basic structure (boilerplate code) and to understand how the app works**

Task-3: Adding Features with AI Assistance Prompt:

Enhance this mobile app by adding: Navigation bar(Home,about,contact) and a simple user registration form ,Basic styling for mobile screens and add cooments(documentation) for each functon

Just now

Code:

```react/page.tsx
import { useState } from "react";

/**
 * Home page component.
 * Displays a welcome greeting and a name input with dynamic "Hello, [name]" resp
 */
export default function Home() {
  /** The current value of the name input field. */
  const [name, setName] = useState("");

  /** The greeting message shown after the user submits their name. */
  const [greeting, setGreeting] = useState("");

  /**
   * Handles form submission.
   * Validates the name field and sets the greeting message.
   * @param e - The form submit event.
   */
  function handleSubmit(e: React.FormEvent) {
    e.preventDefault();
    if (name.trim()) {
      setGreeting(`Hello, ${name.trim()} 👋`);
    }
  }

  /**
   * Clears the greeting and resets the input field.
   */
  function handleReset() {
    setName("");
```

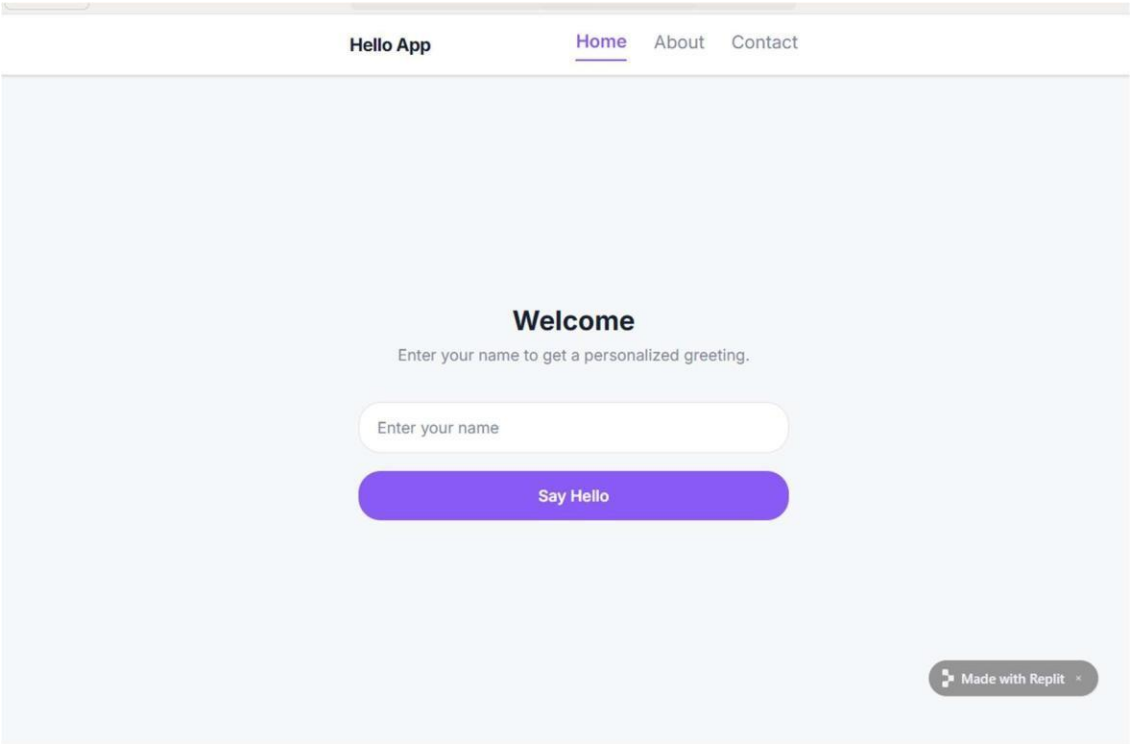
artifacts > mobile-app > src > pages > Contact.tsx > ...

```
import { useState } from "react";

/** Represents the shape of the registration form data. */
type FormData = {
  fullName: string;
  email: string;
  phone: string;
  password: string;
};

/** Represents form field validation errors. */
type FormErrors = Partial<Record<keyof FormData, string>>;

/**
 * Contact page component.
 * Contains a user registration form with client-side validation.
 * On successful submission, shows a confirmation message.
 */
export default function Contact() {
  /** Form field values managed as a single state object. */
  const [form, setForm] = useState<FormData>({
    fullName: "",
    email: "",
    phone: "",
    password: "",
  });

  /** Validation error messages per field. */
  const [errors, setErrors] = useState<FormErrors>({});
```

artifacts > mobile-app > src > pages > About.tsx > ...

```
/**
 * About page component.
 * Displays a brief description of the Hello App and its features.
 */
export default function About() {
  return (
    <main className="flex flex-col items-center min-h-[calc(100vh-56px)] px-6 py-10 bg-backgroun
      <div className="w-full max-w-sm">

        {/* Page heading */}
        <h2 className="text-2xl font-bold text-foreground mb-1">About</h2>
        <p className="text-sm text-muted-foreground mb-8">Learn more about this app.</p>

        {/* App description card */}
        <div className="bg-card border border-border rounded-2xl p-5 mb-4 shadow-sm">
          <h3 className="text-base font-semibold text-foreground mb-2">What is Hello App?</h3>
          <p className="text-sm text-muted-foreground leading-relaxed">
            Hello App is a simple, mobile-friendly web application that demonstrates
            modern React development with clean UI patterns and smooth interactions.
          </p>
        </div>

        {/* Features list card */}
        <div className="bg-card border border-border rounded-2xl p-5 mb-4 shadow-sm">
          <h3 className="text-base font-semibold text-foreground mb-3">Features</h3>
          <ul className="flex flex-col gap-2">
            {[
              "Personalized greeting on the Home page",
              "User registration form with validation",
              "Responsive mobile-first design",
```

Output:

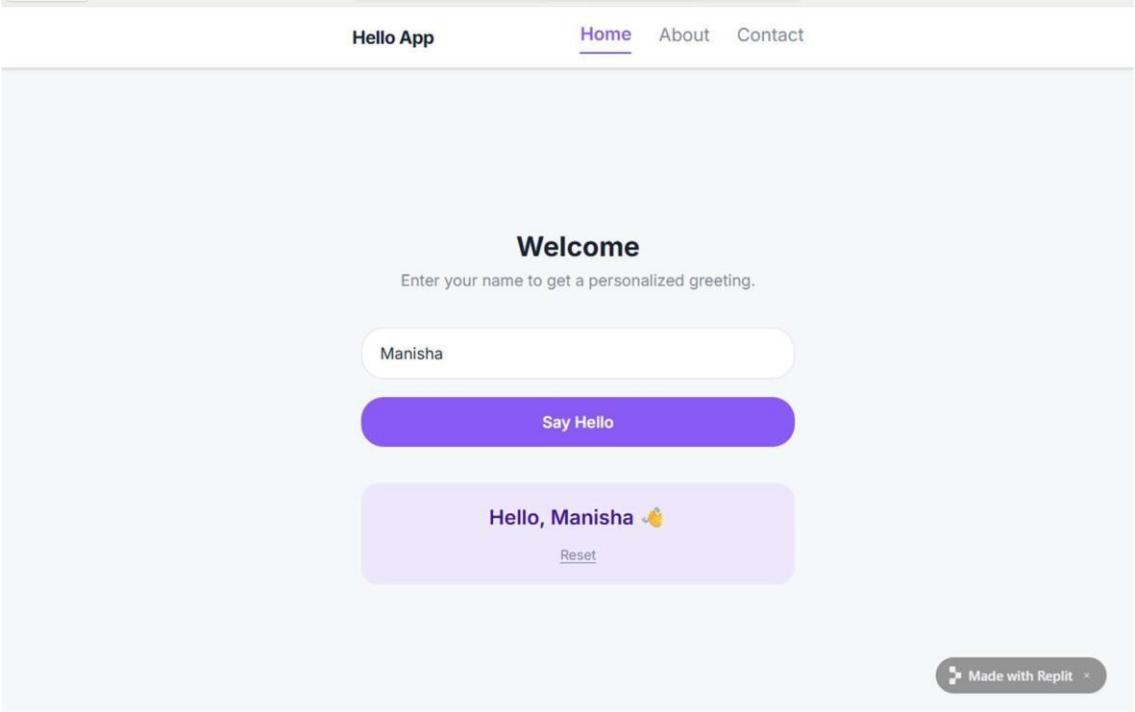


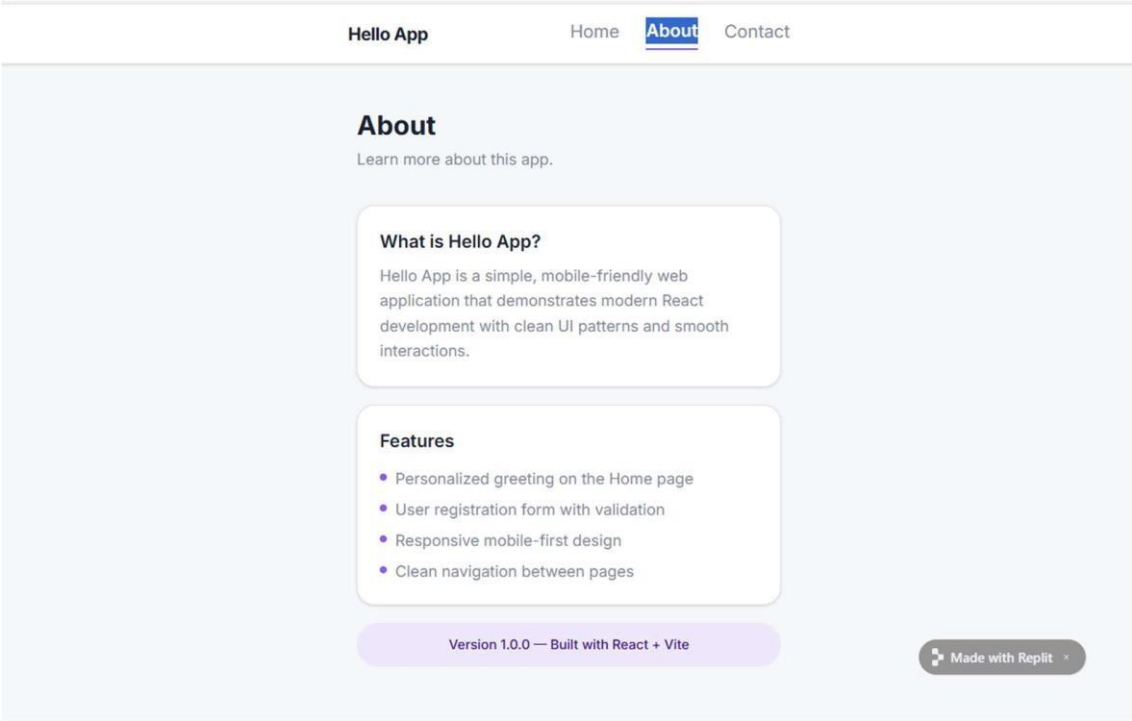
Explanation:

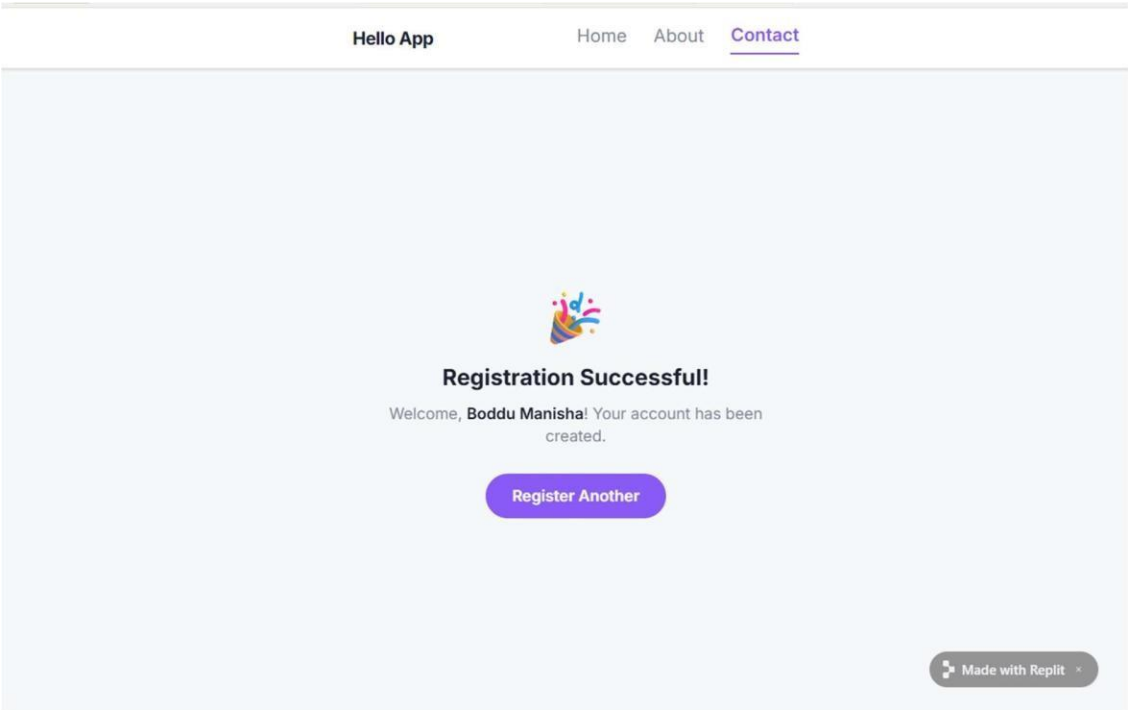
1. **The initial project was created using a Node.js/TypeScript template instead of HTML/CSS/JavaScript.**
2. **Because of this, files like index.html, style.css, and script.js were not available.**
3. **The project structure showed backend files such as package.json and .ts files, which are not suitable for this task.**
4. **To fix this, a new Replit project was created using the HTML, CSS, and JavaScript template.**
5. **This allowed proper access to front-end code and made it easier to build the mobile web application.**

Task-4: Testing on Mobile

Testing:



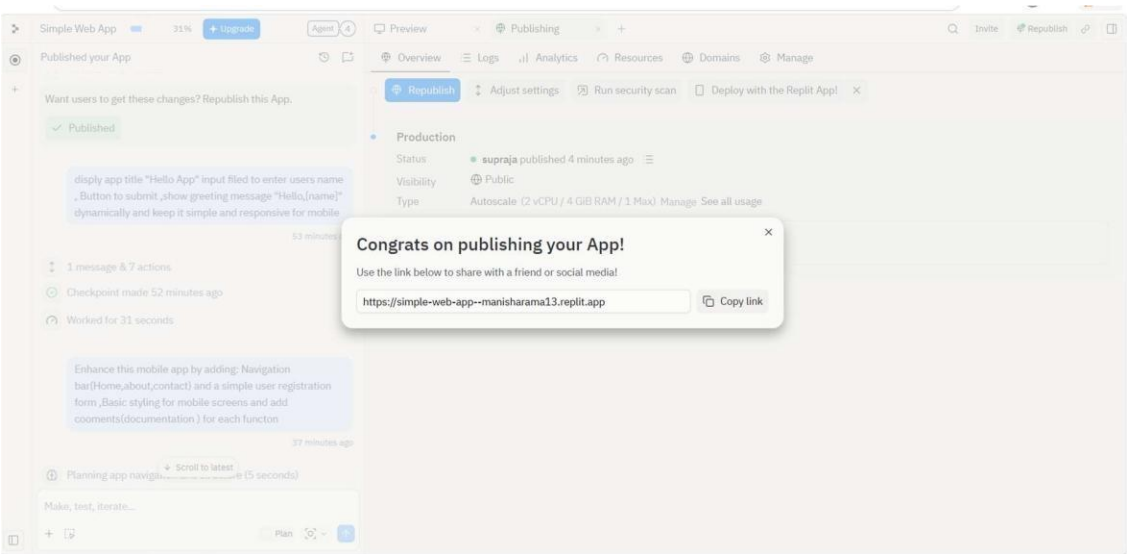

Hello App
Home
About
Contact
About
Learn more about this app.
What is Hello App?
Hello App is a simple, mobile-friendly web application that demonstrates modern React development with clean UI patterns and smooth interactions.
Features
Personalized greeting on the Home page
User registration form with validation
Responsive mobile-first design
Clean navigation between pages
Version 1.0.0 — Built with React + Vite
Made with Replit

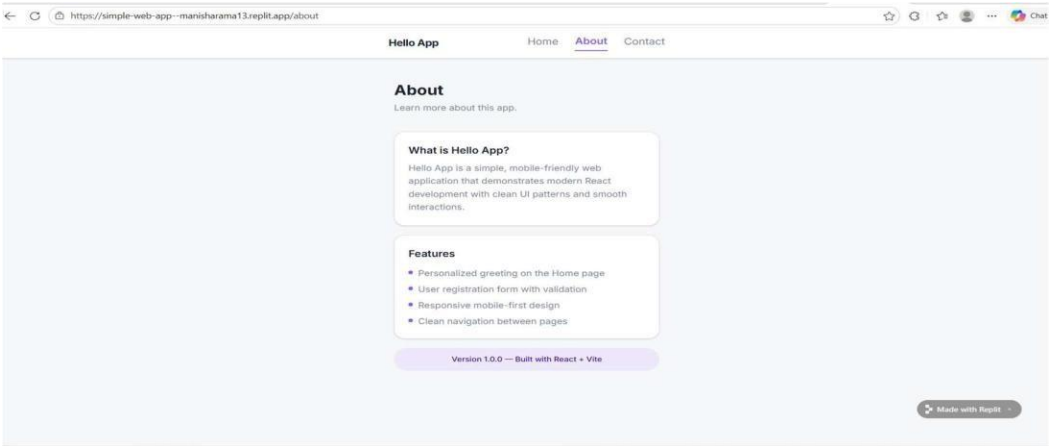

Hello App
Home
About
Contact
Registration Successful!
Welcome, Boddu Manisha! Your account has been created.
Register Another
Made with Replit

Explanation:

1. **The application was run in Replit using the mobile preview mode.**
2. **The app was tested on different screen sizes to ensure proper responsiveness.**
3. **All UI elements like buttons, text, and navigation were checked for alignment.**
4. **Minor UI issues such as spacing and layout were identified and improved.**
5. **AI suggestions were used to fix alignment and enhance the mobile user experience.**

Task-5: Deployment





Explanation:

1. **The application was deployed using Replit's hosting feature.**
2. **The "Publish Now" option was used to generate a public URL for the app.**
3. **The generated link was opened in a browser to test the application.**
4. **The app was successfully accessible online from any device.**
5. **The deployment link was shared with others for testing and feedback.**